

BUDAPEST PESTSZENTLORINC, Hungary

WMO: 128430

Lat: 47.433N

Lon: 19.183E

Elev: 139

StdP: 99.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.8	-7.7	-14.4	1.1	-6.5	-12.1	1.3	-5.7	8.9	5.4	7.8	4.5	1.6	90	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	33.4	20.8	31.6	20.2	29.8	19.5	22.0	29.9	21.1	29.1	20.4	27.9	2.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.5	24.9	18.6	13.7	24.0	17.8	13.0	23.5	64.7	29.8	61.7	29.3	59.2	28.0	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.3	5.4	-12.5	35.7	3.0	2.0	-14.6	37.1	-16.4	38.3	-18.1	39.4	-20.3	40.8		
			-12.9	23.4	2.9	1.0	-15.0	24.1	-16.8	24.7	-18.4	25.2	-20.6	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	0.1	2.1	6.6	12.5	17.0	20.5	22.5	17.0	11.7	6.2	0.8	(6)	
(7)		DBStd	8.88	4.21	4.62	4.22	3.92	3.56	3.54	3.29	3.31	3.56	4.05	4.12	4.25	(7)
(8)		HDD10.0	1109	307	223	118	21	1	0	0	0	29	126	285	(8)	
(9)		HDD18.3	2840	566	455	362	178	69	17	5	6	67	207	364	543	(9)
(10)		CDD10.0	1710	0	1	14	94	217	316	387	377	211	82	11	1	(10)
(11)		CDD18.3	399	0	0	0	2	27	83	133	125	27	1	0	0	(11)
(12)		CDH23.3	3757	0	0	0	34	254	798	1288	1163	210	9	0	0	(12)
(13)		CDH26.7	1352	0	0	0	3	52	278	515	458	46	0	0	0	(13)
(14)	Wind	WSAvg	2.4	2.3	2.6	2.8	2.7	2.6	2.5	2.5	2.3	2.2	2.4	2.3	(14)	
(15)	Precipitation	PrecAvg	547	32	31	31	37	58	63	63	55	47	39	50	39	(15)
(16)		PrecMax	916	90	121	113	103	163	169	183	203	144	150	163	105	(16)
(17)		PrecMin	306	0	0	0	0	3	12	2	0	1	0	0	1	(17)
(18)		PrecStd	129	19	25	21	23	32	35	38	41	35	32	34	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	15.6	20.5	26.6	30.2	34.1	35.7	30.5	24.6	18.2	11.8	(19)	
(20)		MCWB	8.5	9.7	11.6	15.4	18.3	21.0	21.1	20.7	19.1	17.1	13.5	8.7	(20)	
(21)		2%	DB	9.3	12.8	17.8	23.9	28.1	32.0	33.3	33.3	27.8	22.4	15.6	9.4	(21)
(22)		MCWB	6.6	8.2	10.3	14.1	17.7	20.5	20.9	20.5	18.2	15.8	11.9	7.4	(22)	
(23)		5%	DB	7.4	10.6	15.5	21.8	26.3	29.9	31.4	31.2	25.8	20.4	13.7	7.6	(23)
(24)		MCWB	5.5	7.0	9.3	13.1	16.9	19.8	20.2	20.0	17.3	14.8	10.7	5.9	(24)	
(25)		10%	DB	5.8	8.5	13.5	19.8	24.3	27.9	29.6	29.2	23.8	18.2	12.0	6.3	(25)
(26)		MCWB	4.1	6.0	8.2	12.1	16.1	18.8	19.6	19.4	16.4	13.7	9.7	4.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	10.0	12.6	16.3	19.6	22.7	23.0	20.2	17.7	13.7	9.3	(27)	
(28)		MCDWB	11.9	14.5	18.5	24.5	27.8	30.4	31.6	30.7	27.5	23.1	16.8	11.0	(28)	
(29)		2%	WB	7.1	8.5	11.1	14.9	18.5	21.4	22.0	21.7	19.1	16.4	12.4	7.5	(29)
(30)		MCDWB	8.8	11.9	16.2	21.9	25.9	29.2	30.0	29.7	26.0	21.2	15.1	9.0	(30)	
(31)		5%	WB	5.7	7.4	9.9	13.8	17.7	20.6	21.2	20.9	18.1	15.3	11.2	6.3	(31)
(32)		MCDWB	7.2	9.9	14.5	20.2	24.4	28.1	29.0	29.1	24.0	19.3	13.2	7.4	(32)	
(33)		10%	WB	4.2	6.3	8.8	12.7	16.7	19.7	20.4	20.0	17.2	14.2	9.9	5.1	(33)
(34)		MCDWB	5.5	8.4	12.9	18.7	22.8	26.5	27.7	27.7	22.1	17.7	11.7	6.1	(34)	
(35)	Mean Daily Temperature Range	MDBR	5.2	6.7	8.5	10.0	10.3	10.4	10.7	10.9	9.6	8.9	6.2	4.7	(35)	
(36)		5% DB	MCDBR	7.4	9.7	12.2	13.4	13.4	13.2	13.5	13.8	12.7	11.8	8.7	6.1	(36)
(37)		MCWBR	5.1	6.1	6.7	6.0	5.4	4.7	4.3	4.4	5.2	5.9	5.5	4.7	(37)	
(38)		5% WB	MCDBR	7.1	8.6	10.8	12.1	12.1	11.9	11.9	12.3	11.0	10.8	8.0	5.8	(38)
(39)		MCWBR	5.2	5.8	6.4	5.6	5.2	4.7	4.3	4.6	4.9	5.8	5.3	4.6	(39)	
(40)		Clear-Sky Solar Irradiance	taub	0.330	0.353	0.390	0.439	0.428	0.429	0.442	0.441	0.416	0.384	0.359	0.329	(40)
(41)	taud		2.364	2.316	2.244	2.123	2.214	2.244	2.210	2.221	2.258	2.326	2.351	2.372	(41)	
(42)	Ebn at Noon		738	799	821	812	838	838	820	802	785	750	692	690	(42)	
(43)	Edh at Noon		73	93	117	145	137	134	137	130	114	92	74	65	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.04	1.85	3.10	4.57	5.52	5.99	5.87	5.26	3.70	2.37	1.25	0.79	(44)	
(45)		RadStd	0.12	0.27	0.46	0.44	0.57	0.45	0.42	0.51	0.43	0.26	0.12	0.10	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (50 neighbors)	+0.61	N/A	N/A	N/A	+0.55	N/A	-105	-167	+105	+53
		+0.59	N/A	N/A	N/A	+0.54	+0.54	-120	-175	+97	+48

Nomenclature: See separate page